

Ivelina Mladenova

ivelina.p.mladenova@gmail.com | [GitHub](#) | [Website](#) | London, UK

Professional Experience

Quantitative Research Intern — Blue Raven AI, Singapore (Remote)

Oct 2025 – Jun 2026

- Sole researcher working autonomously for 8 months (offered 2-month extension): built an end-to-end equity research pipeline from scratch covering ~6,000 symbols across US, Hong Kong, and Taiwan on 1-minute intraday data. Presented findings in regular reports and near-daily meetings with senior management.
- Engineered 60+ features (momentum, volatility, macro, event-driven, liquidity, options-implied); filtered universe via multi-stage screens on liquidity, market-cap, and composite signal strength.
- Optimised strategy parameters (TP/SL/entry/horizon) initially via random search, then Bayesian optimisation to efficiently narrow the parameter space informed by prior feature calculations; validated signal robustness across market regimes and applied bootstrap significance testing with BH-corrected p-values to guard against overfitting and data-snooping bias.
- Benchmarked high-frequency breakout, momentum, and mean-reversion strategies across multiple horizons; primary momentum-breakout strategy achieved ~28% CAGR over 2020–2024 (4/5 years profitable), post-friction Sharpe of 1.06, Sortino of 1.52, and max drawdown of ~28.8%. Deliberately favoured simple, interpretable signal construction to avoid overfitting; currently stress-testing results and exploring ML-based extensions.

Machine Learning Intern — Earth-i, UK

Jun 2021 – Aug 2021

- Developed an unsupervised change-point detection pipeline for Sentinel-1 (ESA) SAR satellite imagery of copper smelters sourced via Google Earth Engine (TensorFlow): worked with irregularly spaced imagery (~once every 10 days); initially prototyped autoencoders for temporal anomaly detection, then iterated to a simpler, more interpretable approach — normalising and clipping noisy radar data, computing temporal difference layers, and clustering to flag regions of significant pixel-level change, all without labelled ground truth.
- Built an initial framework correlating detected activity spikes with a copper commodity index to test the signal as a leading indicator; achieved ~67% directional accuracy on a binary next-day prediction task, providing early scaffolding for a potential alternative-data trading signal.

Data Analyst (Placement Year) — Sainsbury's Tech, London

Jul 2019 – Jul 2020

- Used SQL and Python (Pandas, statsmodels) to evaluate promotional effectiveness across product categories using hypothesis testing; assessed whether recurring promotions should be re-run, identified optimal timing windows for future campaigns, and reported performance metrics to category buyers and stakeholders.
- Built customer sentiment analysis dashboards in MicroStrategy, giving store managers real-time visibility into feedback trends and enabling faster operational responses.

Finance Analyst (Apprenticeship) — Make Shift, London

Sep 2016 – Sep 2017

- Finance apprentice supporting the accounts manager at a property start-up: handled day-to-day bookkeeping including invoicing, supplier payments, and bank reconciliations. Progressed to preparing monthly P&L and cash-flow reports independently as the role developed.
- Built a dynamic Discounted Cash Flow model in Excel to value a community property investment in Brixton and Peckham ahead of VC fundraising; modelled variables including yearly rent escalations, utility cost changes, occupancy rates, local rental benchmarks, events-driven revenue, and a revenue-share pricing structure for on-site F&B tenants.

Education

PhD in Mathematical Sciences — Queen Mary University of London

Oct 2022 – Present (paused)

Research

- Tackling the curse of dimensionality in volatility-based option pricing (Heston, Wishart extensions) using neural network methods. Developed custom PINN variants with Heston-Black-Scholes PDE loss functions, NTK-based training, and Fourier-based architectures; achieved 10^{-2} pricing accuracy vs. Monte Carlo and Fourier benchmarks, outperforming standard DGM/PINN baselines (~3.14 MSE).
- Improved a high-dimensional Fourier pricing method for Wishart models by deriving optimal damping parameters and bounding the loss function to prevent numerical explosion. Established theoretical approximation results for PINNs on parametric parabolic PDEs. Tools: PyTorch, TensorFlow/Keras; QMUL HPC (Apocrita).

Teaching

- Teaching Assistant across 10 modules — Labs/IT: Probability & Statistics I, Statistical Modelling I, Introduction to Computer Programming (Python), Financial Data Analytics, Introduction to Machine Learning, Numerical Computing with C and C++. Tutorials: Probability & Statistics, Applied Calculus, Introduction to Analysis with Calculus.

Leadership & Outreach

- Led a team of PhD and undergraduate students in a History of Mathematics project, producing a chapter on the mathematical development of probability theory (Law of Large Numbers, Central Limit Theorem) with Python applications in Monte Carlo methods, stochastic processes, and machine learning.
- Organised and led a Women in STEM Hackathon (March 2026): secured funding, coordinated speaker programme, and delivered a talk on "Vibe-Coding Tools" — all while managing a concurrent full-time internship.
- Elected Postgraduate Research Representative; led peer wellbeing initiatives, mentored Year 12 students, and advised researchers across departments (Geography, Biology, Psychology) via a Statistics & Data Science Clinic — providing one-on-one support on statistical analysis, hypothesis testing, and modelling techniques.
- Delivered a Halloween QUIPS talk (internal seminar series) modelling a zombie apocalypse in the PhD office using SIR-type compartmental models to estimate survival odds.

MSc in Applied Statistics — Imperial College London (Merit)

Oct 2021 – Sep 2022

- Thesis: Nowcasted BoE interest rate decisions using a graph-based time-series model on high-dimensional macroeconomic data (with HSBC). Elected Modules: Advanced Simulation, Deep Learning, ML, Statistical Finance, Stochastic Processes.

BSc in Mathematics with Statistics — University of Surrey (First Class)

Oct 2017 – Sep 2021

- Thesis: Numerical estimation via the EM algorithm for parameter inference in discrete-time self-exciting Hawkes processes. Elected Modules: Quantum Mechanics, Mathematics of Data Science, Bayesian Machine Learning and Experimental Design, Time-Series Analysis, Dynamic Asset Pricing.
- Treasurer of MathSoc and Team Surrey Volleyball Club.
- Organised a seminar series on ML and Deep Learning for undergraduate students, coordinating PhD students and lecturers (internal and external) as speakers (Dec 2020 – Apr 2021).

AAT Level 3 Diploma in Accounting — StepForward Apprenticeship (Merit), Sep 2016 – Sep 2017

Technical Skills

Languages: Python (NumPy, Pandas, SciPy, PyTorch, TensorFlow/Keras, Scikit-learn, statsmodels), SQL, Git. **Methods:** Time-series analysis, backtesting, feature engineering, portfolio construction, statistical hypothesis testing; PINNs, DeepONets, CNNs, VAEs; Gaussian Processes, HMMs, MCMC.

Interests

Marathon running, weightlifting, volleyball, rowing, and yoga.